



ASN unit-5 - Adhoc sensor networks notes

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Unit-5

Upper layer issue of WSN:-

→ In wireless sensor networks and ad hoc network the upper layers (application, transport, and sometimes network layers) face specific challenges due to the inherent constraints of these networks, such as limited resources, dynamic topology and energy efficiency.

Transport layer:-

→ The transport layer in Ad hoc and sensor network plays a crucial role in enabling reliable communication b/w nodes.

→ These networks are typically characterized by dynamic topologies, energy constraints and limited resources, so transport layer protocols must address these challenges effectively.

Key functions:-

✓ End-to-End Communications:-

→ Ensures reliable data transfer b/w source and destination nodes.

2. Flow Control:- Manages data flow to avoid congestion and buffer overflow in nodes.

3. Error Control:- Detects and corrects errors in data transmission to ensure accurate delivery.

4. Congestion Control:- Prevents network congestion by controlling data rates.

* Challenges:-

1. Dynamic Topology:- Nodes frequently join and leave the network, affecting routing paths.

2. Limited Resources:- Nodes have limited bandwidth, processing power, and battery life.

3. Scalability:- The network may scale with a large number of nodes, increasing complexity.

4. Wireless Nature:- Signal interference, fading and loss are common in wireless communications.

* Transport layer protocols:-

1. TCP (Transmission Control Protocol):-

• Challenges:- In standard network, TCP provides reliable end-to-end communication.

→ However, in ASN, TCP may perform poorly due to

high packet loss, frequent disconnections and changing topologies.

- Modification:- Protocols like TCP-ELN (Explicit link notification) and TCP for mobile Ad Hoc networks (TCP-MA) modify TCP to handle these issues.

2. UDP (User Datagram Protocol):-

- Advantages:- Lightweight and faster, but it lacks reliability and Congestion Control.
- Suitability:- Often used in real-time applications where timely delivery is more important than reliable delivery.

3. Specialized Protocols:-

- TORA (Temporally Ordered Routing Algorithm) and AODV (Ad hoc On-demand Distance Vector):
→ These protocols help establish efficient routing in ad hoc networks, providing a basis for transport protocols.
- E2ED (End-to-End Delay-based Transport):
→ A transport protocol designed for sensor networks that uses end-to-end delay for flow and congestion control.

4. Energy-Efficient Transport Protocols:

→ These are essential in sensor networks, where battery power is limited. Protocols like EA-TCP (Energy-Aware TCP) aim to reduce energy.

* Key Considerations for Transport Layer Design:-

1. Reliability:-

→ Ensuring reliable data delivery despite frequent link failures.

2. Energy Efficiency:-

→ Minimizing energy consumption to prolong network lifetime.

3. QoS Support:-

→ Providing different levels of service to different applications.

4. Security:-

→ Protecting data from unauthorized access and modification.

High level application layer support:-

- High-level application layer support in ASN focuses on enabling effective Communication, data Processing and resource optimization for applications in these n/w.
- Since these networks are characterized by their decentralized nature, constrained resources and dynamic topologies.
- The application layer plays a critical role in ensuring seamless interaction b/w users and the underlying network.

* key aspects:-

1. Data Aggregation and Dissemination:-

- Purpose: Minimize redundancy and Conserve resources.
- Mechanisms:- Data fusion and aggregation techniques merge data from multiple sensors.
- ↳ Dissemination protocols distribute data efficiently across the n/w.

2. Middleware Services:-

- Role:- simplify application development by abstracting n/w complexities.

• Examples:-

- Service discovery Protocols
- Data-Centric storage
- Event-driven mechanisms.

3. Query Processing:-

- Applications often use query-driven models for accessing data in sensor n/w.
- In-network query Processing reduces overhead by Processing queries locally rather than at a central server.
- Query types include range queries, aggregate queries, and historical queries.

4. Energy Efficiency:-

- Ad hoc and Sensor networks are energy-constrained, so the application layer incorporates energy-saving mechanisms like:

- ↳ Adaptive sampling rates
- ↳ Priority-based data forwarding
- ↳ Efficient encoding and Compression.

5. Scalability and Adaptability

- High-level support ensures applications adapt to changes in n/w size, topology, and density.

→ Examples:

- ↳ Self-Organizing Protocols to manage mobility & node failure.

→ Dynamic reconfiguration of application Parameters

6. QoS (Quality of service)

• Application layer protocols often implement QoS Policies to ensure timely and reliable data delivery.

• Key Parameters:-

↳ latency

↳ Bandwidth

↳ Packet loss rates

7. Security and Privacy:-

→ Application-level encryption and authentication mechanisms protect data from eavesdropping and tampering.

→ Lightweight Protocols address the resource Constraints of ad hoc and sensor nodes.

8. Support for specialized Applications:-

→ The application layer often supports domain-specific functionality for tasks like:

↳ Environmental monitoring

↳ Military surveillance

↳ Health care

↳ Disaster management and search-and-rescue

* Protocol Examples

1. SPIN (Sensor protocols for information via Negotiation):
→ focuses on data-centric dissemination using metadata to minimize redundancy.

2. COUGAR: Supports in-nlw query Processing for efficient data retrieval.

3. Directed Diffusion: Data-centric routing and application layer Protocol for Sensor nlw.

4. TinyDB: A query Processing system for extracting data from Sensor nlw.

Adapting to the inherent dynamic nature of nlw

→ Adapting to the inherent dynamic nature of wireless sensor networks requires designing systems and protocols that handle challenges like topology changes, node mobility, energy Constraints and unpredictable environmental factors.

* key strategies and mechanisms:-

1. Dynamic Topology Management:-

• Challenge:- Frequent changes in nlw topology due to node mobility or failure.

• Adaptation Mechanisms:-

- Proactive Protocols: Maintain routing tables for quick adaptation.
- Reactive Protocols: Discover routes on demand to reduce overhead.
- Hybrid Protocols: Combine Proactive and reactive approaches for flexibility.
- Self-organizing and self-Healing Capabilities:-
- Networks should autonomously adapt to failures or changes without Central Control.

Techniques:

- clustering algorithms for local decision-making and resource management.
- Redundant deployment of nodes to ensure n/w coverage and connectivity.
- Dynamic node role assignment.

3. Energy-Aware Mechanisms:-

challenge:- limited battery life of nodes.

Adaptation Mechanisms:-

- Energy-efficient routing Protocols.
- Duty-cycling to turn off unnecessary sensors or transceivers.

→ load balancing to prevent overuse of specific nodes.

4. Mobility Support:-

• challenge:- node mobility in applications like vehicular ad hoc networks (VANETs) or mobile sensors.

• Adaptation Mechanisms:

- Mobility Prediction algorithm to anticipate movement and adjust routing.
- Use of geographic Routing Protocols to leverage location information.

5. Fault Tolerance and Resilience:-

• challenge:- Node and link failures due to environmental conditions or hardware issues.

• Adaptation Mechanisms

- Redundant Paths for reliable data delivery.
- Error detection and correction mechanisms at the application and transport layers.
- Dynamic reconfiguration to bypass failed nodes.

6. Data Aggregation and Fusion:-

- Challenge: Redundant data transmission increases overhead and energy consumption.

• Adaptation Mechanisms:-

- Aggregation algorithm to combine data from multiple sensors.
- Distributed data fusion techniques to process data locally.

7. Scalable Communication Protocols:

- Challenge: Handling varying n/w sizes and densities.

• Adaptation Mechanisms:-

- Hierarchical routing.
- Adaptive MAC Protocols to adjust to traffic density.

8. Real-Time Data Processing:-

- Challenge: Time-sensitive applications.

• Adaptation Mechanisms:-

- Prioritization of critical data.
- Lightweight QoS mechanisms to ensure low latency and high reliability.

9. Security and Privacy:-

- Challenge: Dynamic environments increase the risk of attacks.

• Adaptation Mechanisms:

- Lightweight cryptographic methods for resource-constrained devices.
- Dynamic trust models to handle node behavior changes.
- Intrusion detection systems that adapt to topology changes.

* Example protocols and Algorithms.

1. LEACH (Low-Energy Adaptive Clustering Hierarchy):

- Adaptive clustering to optimize energy use.

2. AODV (Ad hoc on-Demand Distance Vector Routing):

- On-demand route discovery for dynamic n/w.

3. Directed Diffusion.

- Data-centric communication, adapting to changing interest patterns.

4. Geographic Routing:-

- Location-based routing for dynamic topology.

Sensor networks and mobile robots:-

- The integration of sensor networks and mobile robots creates a dynamic system capable of performing tasks such as environmental monitoring, search and rescue, and autonomous exploration.
- This combination leverages the sensing capabilities of wireless sensor networks and the mobility of robots to overcome challenges posed by static networks or fixed infrastructures.

* Role of Sensor network and Mobile Robots:-

1. Sensor Networks:-

- Composed of static or semi-static sensor nodes.
- Functions: sensing environmental parameters.
 - ↳ Communicating data through multi-hop wireless routing.
 - ↳ Providing situational awareness over large areas.

Mobile Robots:

Equipped with mobility, computation, and communication capabilities.

Functions:

- Act as mobile data collectors reducing energy consumption of static nodes.
- Deploy additional sensor nodes in areas of interest.
- Perform actions based on data collected.

* Key Features.

1. Dynamic Coverage:

- Mobile robots enhance coverage by moving to areas not accessible to static sensors.
- Example: in disaster scenarios, robots can explore collapsed areas while maintaining communication with the static network.

2. Data Relaying:-

- Mobile robots can act as relay nodes, bridging communication gaps in the network.
- Useful in scenarios where static nodes are sparsely distributed or out of range.

3. Energy Optimization:-

- Sensor nodes conserve energy by offloading data transmission tasks to mobile robots.

→ Robots can move closer to sensors, reducing the need for long-range transmissions.

4. Collaborative Sensing:-

→ Robots equipped with sensors can complement the data collected by static nodes.

→ Example:- A robot's camera can validate or refine data from a temperature sensor.

* Benefits:-

1. Flexibility:-

→ Robots extend the capabilities of static sensor n/w to dynamic and large-scale environment.

2. Scalability:-

→ Mobile robots adapt to changes in n/w size and topology.

3. Resilience:-

→ Combined systems are more robust to node failures and environmental challenges.

* Challenges:-

1. Communication and Coordination:-

→ Establishing reliable communication b/w static sensors and mobile robots in a dynamic environment.

→ Maintaining synchronization b/w multiple robots and the sensor n/w.

2. Energy Constraints:-

→ Sensor nodes have limited battery life, necessitating energy-efficient protocols.

→ Robots must optimize their path and tasks to conserve their own energy and the n/w resources.

3. Localization:-

→ Accurate localization is critical for both sensor nodes and mobile robots.

→ Challenges arise in GPS-denied environments.

4. Scalability:-

→ Managing large n/w with multiple robots and sensor nodes.

→ Ensuring seamless operation as the number of nodes and robots increases.

* Key Technologies and Protocols:-

1. localization and navigation
2. Energy-Efficient Communication
3. Path Planning
4. Data Aggregation and Processing.

* Applications:-

1. Environmental Monitoring
2. Search and Rescue
3. Military Surveillance
4. Precision Agriculture
5. Smart Cities.

* Examples.

1. Robomote
2. ANSwarm (Ad hoc N/w Swarm)
3. MARBLE (mobile Autonomous Robot for Biased location Exploration).